1.21 Inequalities: Identifying Regions

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Inequality Regions

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On graph paper, shade the region that satisfies the inequalities:
$2y + x < 14$
$y \geq x^2 - 3x - 4$

You did this at GCSE, the only difference here being that the graphs involved might not be straight lines.

Step 2:
An inequality involving x and y represents a 2D region in space. Identify the correct side of each line each inequality represents.

Click to sketch >

Tip: To quickly sketch $2y + x = 14$, consider what happens when x is 0 and when y is 0.

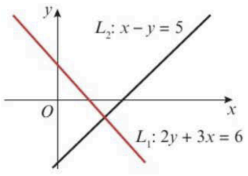
Tip: Make sure y is on the side where it is positive.
If y is on the smaller side, you're below the line.
If y is on the greater side, you're above the line.

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1 L_1 has equation $2y + 3x = 6$. L_2 has the equation $x - y = 5$. The diagram shows a sketch of L_1, L_2.
a Find the coordinates of P, the point of intersection.
b Hence write down the solution to the inequality $2y + 3x > x - y$.



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3 Find the set of values of x for which the curve with equation $y = f(x)$ is below the line with equation $y = g(x)$.

a $f(x) = 3x^2 - 2x - 1$, $g(x) = x + 5$

b $f(x) = 2x^2 - 4x + 1$, $g(x) = 3x - 2$

c $f(x) = 5x - 2x^2 - 4$, $g(x) = -2x - 1$

